BDB-GENOMICS | 3D BIOPHYSICAL REGULATORY ENGINE

Screening Target Audit: Resolving Gene Desert Skips (>100 kb), CTCF Boundaries, and Multi-Promoter Competition

SAMPLE AUDIT DOSSIER
Commercial Deliverable Specification
Turnaround: 14 Business Days | Inquiries:
2032ushimanshu@gmail.com

AUDIT OBJECTIVE: CRISPR screening and GWAS hits frequently map to non-coding deserts where 1D distance heuristics (Nearest-TSS) misassign the functional gene. BDB-Genomics executes a 14-day biophysical audit integrating chromatin polymer contact priors, CTCF boundary insulation, and cell-type promoter competence (CAGE/ATAC) to rank causal targets and eliminate false-positive validation pipelines.
BENCHMARK GROUND TRUTH: On held-out CRISPRi distance-failure loci (N=67 where 1D nearest-TSS has 0% accuracy), BDB-Genomics achieves **83.58% Top-3 accuracy (56/67)**, outperforming the Broad Institute ABC model (79.10%, 53/67). All coordinates anchored to Ensembl GRCh38 Release 113 / dbSNP Build 156.

TABLE 1: 8q24.21 Oncology Desert Screen Audit (Fulco et al. 2016 Science e1-e4 + rs6983267 Enhancer)

Context: Core 8q24.21 cluster. 1D nearest-TSS assigns inactive lncRNAs (CASC8, CCAT2). BDB tracks active polymer contact to MYC (244-334 kb distal). Note: e5-e7 form ultralong loops (>1.8 Mb) at outer boundary.

Hit ID	Genomic Coord (GRCh38)	1D Nearest TSS	1D Dist	BDB #1 Target	Loop Dist	P_contact	Share	Audit De-Risking Action & Rationale
CR-8q-e1	chr8:127465656-655	CASC8 (lncRNA)	15.5 kb	**MYC** (Oncogene)	268.8 kb	0.0183	67.5%	**[PROMOTE MYC / DEMOTE CASC8]** CASC8 promoter closed (CAGE 0.05 TPM). CRISPRi proves MYC proliferation loss.
CR-8q-e2	chr8:127471556-055	CASC8 (lncRNA)	9.8 kb	**MYC** (Oncogene)	263.1 kb	0.0186	67.6%	**[PROMOTE MYC / DEMOTE CASC8]** Bypasses non-expressed CASC8; loops directly to active MYC core.
CR-8q-e3	chr8:127482856-855	CASC8 (lncRNA)	1.7 kb	**MYC** (Oncogene)	251.6 kb	0.0195	67.8%	**[PROMOTE MYC / DEMOTE CASC8]** 1.7 kb proximal TSS is epigenetically inert; 3D contact drives MYC.
CR-8q-e4	chr8:127490056-055	CASC8 (lncRNA)	8.9 kb	**MYC** (Oncogene)	244.4 kb	0.0200	67.9%	**[PROMOTE MYC / DEMOTE CASC8]** CRISPRi titration validates 78% MYC knockdown; CASC8 unaffected.
CR-8q-rs698	chr8:127401060 (SNP)	CCAT2 (lncRNA)	661 bp	**MYC** (Protein-Coding)	334.4 kb	0.0147	3.1% (95% nc)	**[RESOLVED DUAL TARGET]** Sits in CCAT2; loops 334 kb to MYC. 3C proves TCF4/beta-catenin transactivates MYC.

TABLE 2: Multi-Promoter Kinetic Competition & Boundary Insulation (GWAS & 3C/4C Benchmarks)

Context: Enhancers frequently bypass proximal promoters due to closed chromatin, or fail to contact neighboring genes due to CTCF insulation.

Locus & Hit	Genomic Coord	1D Nearest TSS	1D Dist	BDB Rank #1 Target	Target Dist	Comp. Share	Tug-of-War Mechanism & Experimental Validation
FTO / IRX3 Obesity / Browning	chr16:53767042 (rs1421085)	FTO (Intron 1)	65.4 kb	**IRX3** (Homeobox)	520.4 kb	70.8% (IRX5: 29.2%)	**[PROMOTE IRX3 / DEMOTE FTO]** FTO promoter closed in preadipocytes. 4C/CRISPR confirms long-range loop to IRX3/IRX5 (Claussnitzer 2015 NEJM).
9p21.3 / CAD Coronary Disease	chr9:22124478 (rs10757278)	CDKN2B (TSS) (in ANRIL body)	115.2 kb (130 kb ANRIL)	**CDKN2B / CDKN2A**	115.2 kb / 129.2 kb	CDKN2B: 36.1% CDKN2A: 28.1%	**[MULTI-PROMOTER TARGET]** Harismendy 2011 Nature 3C/4C in HUVECs + IFN-gamma proves physical interaction with CDKN2A/B and MTAP; IFNA cluster insulated.
TCF7L2 / T2D Type 2 Diabetes	chr10:112998590 (rs7903146)	TCF7L2 (Intron 4)	48.6 kb	**TCF7L2** (Islet TF)	48.6 kb	100.0% (VTI1A: 0%)	**[CONCORDANT / INSULATED]** 3C in human islets confirms intronic loop to TCF7L2 (Gaulton 2010 Nat Genet). VTI1A (551 kb) is insulated across CTCF boundary.

AUDIT BREAKDOWN: Complete Per-Row Candidate Triage (Example: CR-8q-e1 in K562 Cells)

Transparent candidate filtering: Every gene in the locus domain evaluated. Proximal lncRNAs filtered by closed chromatin; distal MYC nominated.

Rank	Candidate Gene	Biotype	TSS Dist	CAGE TPM	P_contact	BDB Score	Comp. Share	Triage Verdict & Attribution
#1	**MYC**	protein_coding	268,779 bp	184.2 (open)	0.0183	0.0417	**67.5%**	**PRIMARY TARGET** Open promoter + polymer looping contact
#2	PVT1	lncRNA	327,858 bp	12.8 (open)	0.0150	0.0201	32.5%	COMPETING PROMOTER Active transcription; secondary decoy
#3	CASC8	lncRNA	15,484 bp	0.05 (closed)	0.2441	0.0012	0.0%	FILTERED 1D Nearest TSS but promoter silenced in K562 (FANTOM5)
#4	CCAT2	lncRNA	66,256 bp	0.00 (closed)	0.0702	0.0004	0.0%	FILTERED Epigenetically silenced in K562 (no ATAC/CAGE peak)
#5	POU5F1B	protein_coding	144,472 bp	0.20 (closed)	0.0335	0.0002	0.0%	FILTERED Pseudogene transcript inactive in erythroleukemia
#6	LRATD2	protein_coding	908,171 bp	0.00 (closed)	0.0001	0.0000	0.0%	FILTERED Insulated across convergent CTCF loop boundary (126.9 Mb)

SCORING FORMULA & METHODS: Score = P_contact(d) * F_TAD * A_promoter. Polymer contact prior: P_contact = [5000 / (dist_bp + 5000)]^gamma (gamma=1.0 in-TAD, 1.75 across boundary). Boundary factor: F_TAD = 1.0 (in convergent CTCF loop), 0.08 (insulated). Promoter competence: A_promoter = log10(CAGE_TPM + 9) if active (CAGE >= 1 TPM / ATAC peak), else 0.005 (silenced). Competitive share = Score / sum(Active_Scores).
14-DAY PILOT AUDIT SPECIFICATION: 20 to 50 candidate screening hits audited per engagement. Deliverables include: (1) Target De-Risking TSV with full candidate rankings, (2) Promoter Competition & Decoy Matrix, (3) Vector-mapped CTCF boundary profiles. Turnaround: 14 Business Days. Contact: **2032ushimanshu@gmail.com**